

Responses to the comments by Reviewer 2:

General Response: Thank you very much for your positive feedback on our revised manuscript. In response to your valuable comments, we have enriched the revised manuscript with new data and expanded discussions. These enhancements are highlighted in pink for easy identification. Your insightful observations have significantly contributed to improving the manuscript. We would greatly appreciate your review of the revised manuscript at your convenience. We are open to and welcome any additional insights you may have. Once again, we are profoundly grateful for your meticulous review of our manuscript.

Comments:

The authors have addressed most of my previous concerns. In particular, additional data regarding ALD thickness optimization and the discussion of the unique advantages of ALD have significantly strengthened the manuscript. The revised version provides a much clearer reason for employing ALD-derived ZnF_2 coatings. Overall, the manuscript has been substantially improved and is now much closer to the standard expected for publication in Advanced Energy Materials. After the following comments are adequately addressed, the manuscript can be considered for publication.

1. The authors should provide a schematic illustration of the evolved NaF/Na–Zn interphase after the conversion of ZnF_2 . The schematic should clearly show the spatial distribution of the Na–Zn alloy and NaF components and explicitly illustrate their respective roles in reducing the nucleation overpotential, promoting uniform Na deposition, stabilizing the SEI, and regulating Na^+ flux.

Response: Thank you for your insightful comment. In accordance with your suggestion, we have added a schematic illustration to clarify the mechanism responsible for the enhanced reversibility of Na plating/stripping on the ZnF_2/Al . The schematic explicitly depicts the spatial distribution of the Na–Zn alloy and NaF components, as well as their respective roles in lowering the Na nucleation overpotential, promoting uniform Na deposition, stabilizing the SEI, and regulating Na^+ flux.

To further elucidate the origin of the improved Na plating/stripping reversibility on the ZnF_2/Al , the schematic illustration and the corresponding discussion have been incorporated into the revised manuscript as follows:

On page 17:

“... Taken together, these results strongly support our hypothesis. In contrast to the poor reversibility of Na plating/stripping on bare Al, the ALD-engineered ZnF_2/Al current collector undergoes an *in situ* transformation into a NaF/Na–Zn interphase that facilitates uniform Na nucleation, regulates Na-ion flux, suppresses dendritic growth, and significantly improves the reversibility of Na plating and stripping (Figure S21). As a result, enhanced electrochemical performance is achieved in both lean-Na and anode-free full-cell configurations. ...”

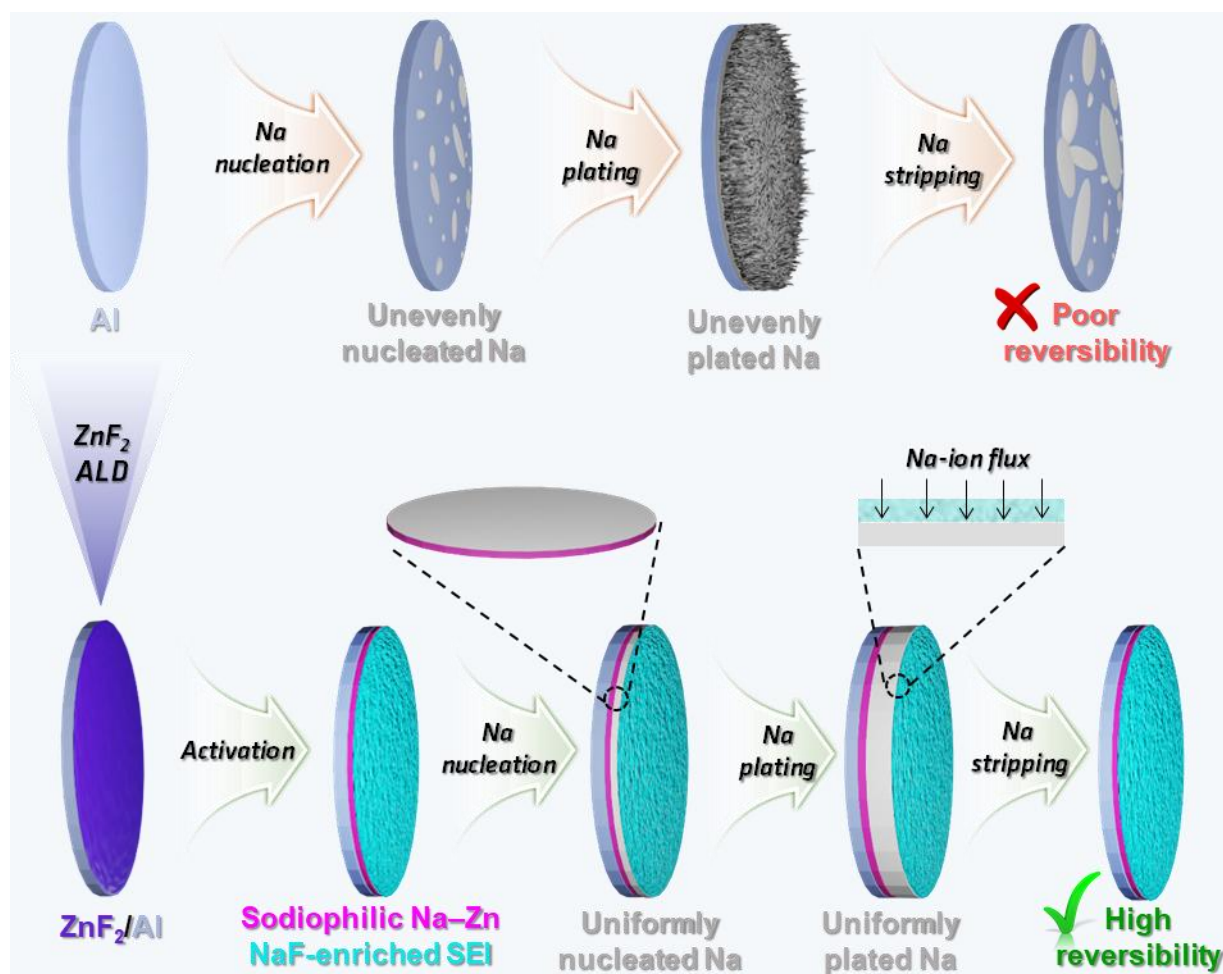


Figure S22. Schematic illustration of the mechanism underlying the enhanced reversibility of Na plating/stripping on the ZnF_2/Al CC.

2. Additional Na nucleation measurements at 0.1 and 1 mA cm^{-2} are recommended to more comprehensively demonstrate the advantages of ZnF_2/Al under different current densities. Furthermore, CV measurements at multiple scan rates (at least 3 scan rates) should be provided to further support the kinetic analysis.

Response: Thank you for your constructive comment. In response to your suggestion, we

have included additional Na nucleation measurements at 0.1 and 1.0 mA cm⁻², as well as CV data collected at multiple scan rates, in the revised manuscript.

Encouragingly, ZnF₂/Al consistently exhibited significantly lower Na nucleation overpotentials than bare Al across a wide range of current densities (0.1, 0.5, 1.0, and 2.0 mA cm⁻²). This improved electrochemical behavior can be attributed to the enhanced sodiophilicity imparted by the Na–Zn alloy–containing interphase. Furthermore, the CV curves obtained at various scan rates consistently showed higher peak current densities and lower polarization potentials for the Na/ZnF₂/Al||NVP cell, indicating faster reaction kinetics and improved charge-transfer characteristics of the Na/ZnF₂/Al anode.

Accordingly, the additional Na nucleation and CV data, together with the corresponding discussion, have been incorporated into the revised manuscript as follows:

On page 11:

“... Moreover, across a wide range of current densities (0.1, 1.0, and 2.0 mA cm⁻²), ZnF₂/Al consistently exhibited significantly lower overpotentials than bare Al (Figures S10–S11). This enhanced electrochemical behavior is likely attributable to the improved sodiophilicity imparted by the Na–Zn alloy–containing interphase. ...”

On page 16:

“... Similarly, cyclic voltammetry (CV) curves under various scan rates consistently revealed a higher peak current density and a lower polarization potential of the Na/ZnF₂/Al||NVP cell (Figure S16), indicating the better reaction kinetics of the Na/ZnF₂/Al anode. ...”

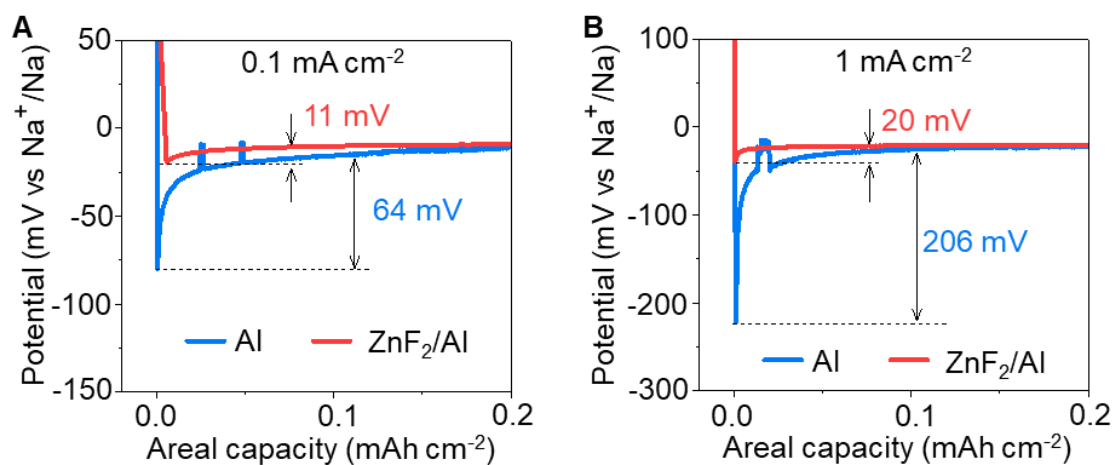


Figure S10. Overpotential of Na nucleation on bare Al and ZnF₂/Al. A) At a current density of 0.1 mA cm⁻². **B)** At a current density of 1.0 mA cm⁻².

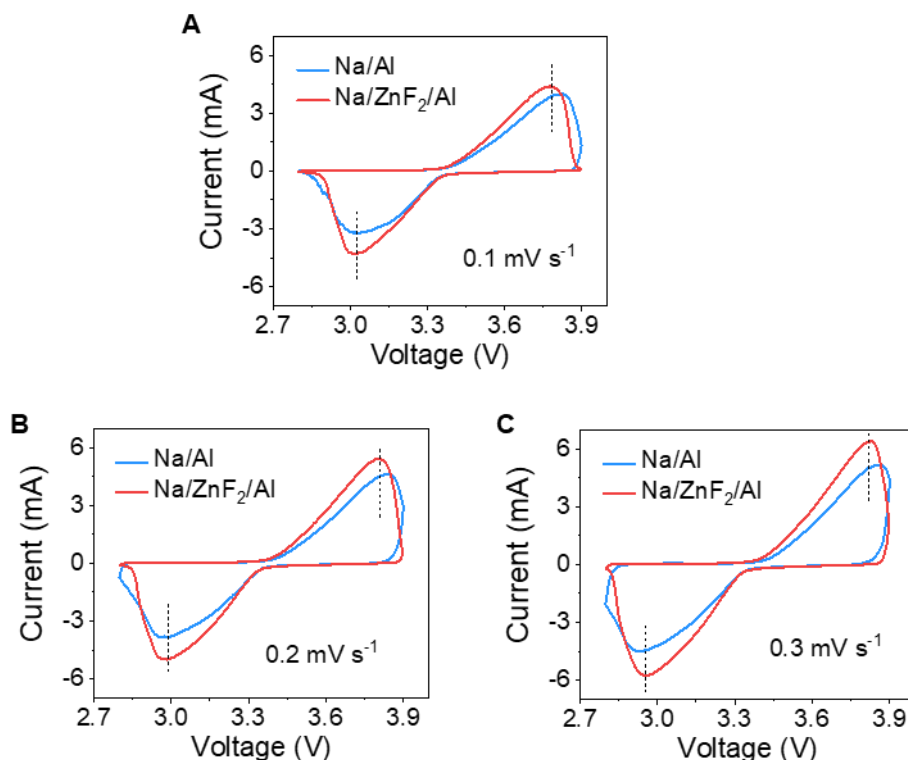


Figure S17. CV curves of the Na/Al || NVP and Na/ZnF₂/Al || NVP lean-Na cells. Comparison of CV curves under scan rates of (A) 0.1, (B) 0.2, and (C) 0.3 mV s⁻¹.

3. The detailed methodology, equations, and parameters used for the energy-density calculations should be provided in the Supporting Information.

Response: Thank you for your valuable comment. We have included the information regarding the energy-density calculations in the Supporting Information as follows:

On page 6, Supporting Information:

“ Calculations of energy density

The gravimetric density is calculated by the following equation:

$$E_g = \frac{U \times C}{\sum m_i}$$

where E_g is the gravimetric energy density (Wh kg⁻¹), U is the average voltage (3.36 V), C is the cell capacity (Ah), and $\sum m_i$ is the total weight of the active materials at the anode and cathode (kg). In our anode-free cells, the mass of ultrathin ZnF₂ is negligible. Therefore, $\sum m_i$ is the weight of the active material at the cathode.”

4. The benchmarking should be expanded to include more recently reported anode-free sodium battery systems to better position the present work within the current state of the art.

Response: Thank you for your insightful comment. In accordance with your suggestion, we have incorporated a table showing a comparison of the cycling performance of anode-free full cells with previously reported studies as follows:

Table 2: Comparison of the cycling performance of anode-free full cells with previously reported studies

No.	Anode	Cathode	Cathode mass loading (mg cm ⁻²)	Current density	Cycling number	Capacity retention (%)	Ref.
1	F-A-Al	NVP	11.3	0.3 C	50	46.1	S11
2	Cu@Au	FeS ₂	2	1.0 A g ⁻¹	50	55	S12
3	AuPC	NVP	3.5	3.0 mA cm ⁻²	100	31	S13
4	Al-Cu	NVP/C	3.5–4.0	1.0 C	50	72	S14
5	FCTD	NVP	7.4	2.0	100	74.7	S15
6	Sn/Al	NVP	18.32	1.0 C	88	~32	S16
7	Cu	NVP	4.0	0.5 mA cm ⁻²	100	~55	S17
8	ZnF ₂ /Al	NVP	12.2	0.1 C	60	55	This work
			14.3	1.0 C	100	62	

(S11) Wu, S.; Hwang, J.; Matsumoto, K.; Hagiwara, R. The Rational Design of Low-Barrier Fluorinated Aluminum Substrates for Anode-Free Sodium Metal Battery. *Adv. Energy Mater.* **2023**, *13*, 2302468.

(12) Tang, S.; Qiu, Z.; Wang, X. Y.; Gu, Y.; Zhang, X. G.; Wang, W. W.; Yan, J. W.; Zheng, M. Sen; Dong, Q. F.; Mao, B. W. A Room-Temperature Sodium Metal Anode Enabled by a Sodiophilic Layer. *Nano Energy* **2018**, *48*, 101–106.

(13) Kim, S.; Ryoo, G.; Lee, J. H.; Kim, J.; Park, J. H.; Cho, K. Horizontal Sodium Growth via the Sodiophilicity-Driven Structural Design of Current Collectors for Anode-Free

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- (14) Li, H.; Zhang, H.; Wu, F.; Zarrabeitia, M.; Geiger, D.; Kaiser, U.; Varzi, A.; Passerini, S. Sodiophilic Current Collectors Based on MOF-Derived Nanocomposites for Anode-Less Na-Metal Batteries. *Adv. Energy Mater.* **2022**, *12*, 2202293.
- (15) Zhuang, R.; Zhang, X.; Qu, C.; Xu, X.; Yang, J.; Ye, Q.; Liu, Z.; Kaskel, S.; Xu, F.; Wang, H. Fluorinated Porous Frameworks Enable Robust Anodeless Sodium Metal Batteries. *Sci. Adv.* **2023**, *9*, eadh8060.
- (16) Ma, P.; Zhang, Y.; Li, W.; Luo, J.; Wen, L.; Tang, G.; Gai, J.; Wang, Q.; Zhao, L.; Ge, J.; Chen, W. Tailoring Alloy-Reaction-Induced Semi-Coherent Interface to Guide Sodium Nucleation and Growth for Long-Term Anode-Less Sodium-Metal Batteries. *Sci. China Mater.* **2024**, *67*, 3648–3657.
- (17) Panchal, R. A.; Datta, J.; Varude, V.; Bhimani, K.; Mahajani, V.; Kamble, M.; Anjan, A.; Manoj, R. M.; Zha, R. H.; Datta, D.; Koratkar, N. Nano-Silica Electrolyte Additive Enables Dendrite Suppression in an Anode-Free Sodium Metal Battery. *Nano Energy* **2024**, *129*, 110010.

On page 16:

“... Consequently, the cycling performance of the $\text{ZnF}_2/\text{Al}||\text{NVP}$ cell is comparable to, or even surpasses, that of previously reported systems, underscoring the effectiveness of the ZnF_2/Al CC (Table S1). ...”

We hope that our revised manuscript meets your expectations. We warmly welcome any further comments you may have. Once again, we express our sincere gratitude for your insightful comments.